JAN-S-627

14 JULY 1948

NATIONAL MILITARY ESTABLISHMENT SPECIFICATION

SOLDER; LOW-MELTING-POINT

This specification was approved by the Departments of the Army, the Navy, and the Air Force for use of procurement services of the respective Departments, and supersedes the following specifications:

U. S. Army
57–98
6 Aug. 1931

Navy Department
Bureau of Ordnance
47S47(Ord)
25 Feb. 1947

A. APPLICABLE SPECIFICATIONS AND OTHER PUBLICATIONS

A–1. *Specifications.*—The following specifications, of the issue in effect on the date of invitation for bids, form a part of this specification:

JOINT ARMY-NAVY SPECIFICATIONS

JAN–P–105—Packaging and Packing for Overseas Shipment—Boxes, Wood, Cleated, Plywood.
JAN–P–106—Packaging and Packing for Overseas Shipment—Boxes, Wood, Nailed.
JAN–B–107—Packaging and Packing for Overseas Shipment—Boxes, Wood, Wirebound (Overseas Type).
JAN–P–108—Packaging and Packing for Overseas Shipment—Boxes, Fiberboard (V-Board and W-Board), Exterior and Interior.
JAN–P–125—Packaging and Packing for Overseas Shipment — Barrier-Materials, Waterproof, Flexible.
JAN–P–138—Packaging and Packing for Overseas Shipment—Boxes, Wood, Fiberboard-Lined.
JAN–P–140—Packaging and Packing for Overseas Shipment—Adhesives, Water-Resistant, Case-Liner.

FEDERAL SPECIFICATIONS

NN–B–591—Boxes; Wood-Cleated-Fiberboard.
NN–B–601—Boxes; Wood-Cleated-Plywood, For Domestic Shipment.
NN–B–621—Boxes; Wood, Nailed and Lock-Corner.
NN–B–631—Boxes; Wood, Wire-Bound, For Domestic Shipment.
QQ–M–151—Metals; General Specification for Inspection of.
LLL–B–631—Boxes; Fiber, Corrugated (For Domestic Shipment).
LLL–B–636—Boxes; Fiber, Solid (For Domestic Shipment).

U. S. ARMY SPECIFICATION

100–2—Standard Specification for Marking Shipments by Contractors.[1]

[1] Applicable only to Army purchases.

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NAVY DEPARTMENT SPECIFICATION

General Specifications for Inspection of Material, and Appendix II, Metals, Part A, Definitions and Tests.[2]

A–2. *Other publication.*—The following publication, of the issue in effect on date of invitation for bids, forms a part of this specification:

BUREAU OF SUPPLIES AND ACCOUNTS PUBLICATION

Navy Shipment Marking Handbook.[2]

B. COMPOSITIONS AND FORMS

B–1. *Compositions.*—This specification covers the following four compositions of low-melting-point solder, as specified (see pars. E–1 and H–2):

Composition A.
Composition B.
Composition C.
Composition D.

B–2. *Forms.*—This specification covers low-melting-point solder in the form of bars, ingots, sheets, strips, and other forms, as specified.

C. MATERIAL AND WORKMANSHIP

C–1. *Material.*—The raw materials shall be such as to produce solders meeting the requirements for the various compositions of this specification.

C–2. *Workmanship.*—Solder shall be uniform in quality and color, and shall be free from any injurious defects which may affect the serviceability. Sheets and strips shall be commercially flat, of uniform thickness, and free from scale, laps, twists, and damaged ends or edges.

D. GENERAL REQUIREMENTS

D–1. See section E.

E. DETAIL REQUIREMENTS

E–1. *Chemical and freezing point requirements.*—Solder shall meet the chemical and freezing point requirements shown in table I.

TABLE I.—*Chemical and freezing point requirements.*

Composition	Tin	Lead	Bismuth	Cadmium	Other elements Total	Freezing-point range
	Percent	*Percent*	*Percent*	*Percent*	*Percent*	*Deg. F.*
A.....	42–44	41–45	13–15	.....	.....	330–285
B.....	16–18	15–17	66–68	.....	.....	305–300
C.....	10.8–13.2	25.2–30.8	45–55	9–11	1.00	165–155
D.....	36–44	18–22	36–44	.....	1.00	220–210

E–1a. An analysis of each lot shall be furnished by the contractor, showing the percentage of each element specified.

[2] Applicable only to Navy purchases.

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E–2. *Dimensions.*—Solder in all forms shall conform to the dimensions specified in the contract or order. (See par. H–2.)

E–3. *Marking.*—Unless otherwise specified, each bar, ingot, sheet, strip, or other form, shall have cast or stamped thereon the specification number, composition, and the contractor's brand or symbol. In the case of wire solder, the marking may be placed on the spool.

F. METHODS OF SAMPLING, INSPECTION, AND TESTS

F–1. *Inspection.*—

F–1a. *Place of inspection.*—Unless otherwise specified, inspection shall be made at the place of manufacture. (See par. H–2.)

F–1b. *Dimensions and marking.*—The inspector shall examine a sufficient number of pieces from each lot to determine compliance with paragraphs E–2 and E–3.

F–2. *Lot.*—A lot shall consist of all solder produced from the same melt, of the same form and size, submitted for inspection at the same time.

F–3. *Sampling for Government analysis.*—

F–3a. *Bars and ingots.*—At least three pieces shall be selected by the Government inspector from a lot of 1,000 pounds or less; at least five pieces from a lot of over 1,000 to 10,000 pounds, inclusive; and at least 10 pieces from a lot of over 10,000 pounds to one carload, for use in preparing the composite sample as described in paragraph F–3c.

F–3b. *Other forms.*—In case the lot is in any other form, an equivalent quantity of metal representing the lot shall be taken, but in no case shall less than one whole piece (except in the case of wire) represent a lot.

F–3c. The sample for chemical analysis shall be composed of sawings or drillings taken from each piece selected to represent the lot in accordance with paragraph F–3a and paragraph F–3b. Sawings or drillings shall be taken so as to represent the average cross section of each piece. The sawings or drillings from all pieces shall be thoroughly mixed, and not less than 5 ounces of the mixture shall be forwarded by the inspector for analysis. Samples from Navy purchases shall be forwarded to a Government laboratory.

F–4. *Tests.*—

F–4a. *Bismuth.*—Transfer an accurately weighed portion of 0.5 gm. of the sample to an 800-ml. beaker and add 50 ml. of water, 5 ml. of 12 N hydrochloric acid and 10 ml. of 15 N nitric acid. Cover the beaker with a watch glass and heat until the sample is completely dissolved. Add 50 ml. of water to the beaker and make the solution just alkaline with a 20 percent potassium hydroxide solution using phenophthalein as an indicator. Add 100 ml. of 20 percent potassium hydroxide solution, dilute the mixture to 500 ml. with water and pass hydrogen sulfide gas into the solution for 15 minutes. Place the beaker and contents on a steam bath for 2 hours, allow the beaker and contents to cool to room temperature and then filter the contents of the beaker with the aid of a No. 42, 11 cm. Whatman or equivalent filter paper. Transfer the residue

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to the filter paper and wash the residue and the filter paper 5 times with approximately 20 ml. portions of a 0.5 percent potassium hydroxide solution saturated with hydrogen sulfide. Open the filter paper and place it on the upper inside wall of the original 800-ml. beaker. Wash most of the precipitate from the paper into the beaker with a fine stream of hot water, using about 50 ml. of water. Dissolve the precipitate remaining on the filter paper with about 2 ml. of 15 N nitric acid solution, added dropwise. Wash the filter paper with about 40 ml. of 0.7 N nitric acid solution and then discard the filter paper. Cover the beaker and boil the contents for about 5 minutes. If the sulfides are not completely dissolved, add 1 ml. of 15 N nitric acid solution and again boil the mixture for about 5 minutes. Repeat the addition of nitric acid and boiling until the sulfides are completely dissolved and the separated sulfur is yellow to white in color. Filter the contents of the beaker with the aid of a No. 42, 9 cm. Whatman or equivalent filter paper and catch the filtrate in a 400-ml. beaker. Transfer the residue to the filter paper and wash 5 times with approximately 10-ml. portions of a 0.7 N nitric acid solution. Neutralize the solution by adding 15 N ammonium hydroxide, dropwise, using litmus paper as an indicator and record the number of ml. of 15 N ammonium hydroxide used.[a] Add 10 ml. of 15 N nitric acid and dilute the solution to 250 ml. with water. Cool the beaker and contents to 18 ± 2°C. Add to the beaker, dropwise, and with constant stirring of the contents, a freshly prepared solution of 5 percent cupferron in the following amounts: For solder, Composition A, use 10 ml. of cupferron solution, for Composition B, use 25 ml, and for Compositions C and D, use 20 ml. Allow the contents of the beaker to settle for 15 minutes and then filter, using a tared ignited asbestos Gooch crucible. Transfer the precipitate to the crucible and wash 5 times with approximately 15-ml. portions of a freshly prepared solution of 0.1 percent cupferron in 0.15 N nitric acid solution. (Reserve the filtrate for the determination of the lead content.) Dry the crucible and contents at 100 ± 10°C. for 30 minutes and then place on a hot plate heated to 225 ± 25°C. for about one hour to convert most of the residue to bismuth oxide. (On heating, the residue effervesces, turns black and eventually assumes the light brown coloration of bismuth oxide.) Heat the crucible and contents at 650 ± 25°C. for one hour, cool and weigh. Repeat this procedure until constant weight is attained. Consider the residue in the crucible to be bismuth oxide and calculate the increase in weight of the crucible to percentage of bismuth in the sample as follows:

$$\text{Bismuth, percent} = \frac{\text{Weight of residue in crucible} \times 89.70}{\text{Weight of sample}}$$

F-4b. *Lead.*—Transfer the filtrate obtained from the bismuth determination to a 600 ml. beaker. Evaporate the solution to about 50 ml. and add 36 N sulfuric acid in the following amounts: 5 ml. plus one-half of the volume of ammonium hydroxide used in connection with the determination of bismuth. (For example, if 12 ml. of ammonium hydroxide were used, use 5 plus ½ times 12 or 11 ml.

[a] This value is used in connection with the determination of the lead content.

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of sulfuric acid.) Heat the contents of the beaker until sulfur trioxide fumes appear. If the residue is colored by the presence of organic matter, add 2 ml. of 15 N nitric acid solution and again heat the mixture until sulfur trioxide fumes are produced. Repeat the addition of the nitric acid solution and then heat to sulfur-trioxide fumes until all the organic matter in the beaker is destroyed and the residue in the beaker is white or colorless. Add 5 ml. of water and again heat the mixture until sulfur trioxide fumes are produced. Add 100 ml. of water, heat the mixture to boiling, and boil for 15 minutes. Allow the mixture to stand at room temperature for at least 3 hours and then filter it with the aid of a tared, previously ignited filtering crucible. Transfer the precipitate to the crucible and wash it 5 times with approximately 10 ml. portions of 1.5 N sulfuric acid solution. Wash the crucible and residue with 3 ml. of water. Dry the crucible as lead sulfate and calculate its weight to percentage of lead in the sample as follows:

$$\text{Lead, percent} = \frac{\text{Weight of residue in crucible} \times 68.32}{\text{Weight of sample}}$$

F-4c. *Tin.*—One-half gram of the sample is placed in a 300-ml. Erlenmeyer flask and 10 ml. of strong sulfuric acid is added. Nearly all of the excess acid is expelled. The moist residue is allowed to cool, and 100 ml. of air-free water is added, followed by 50 ml. of strong hydrochloric acid and the mixture gently warmed until the solution begins to clear. Into the Erlenmeyer flask is placed a 1-hole stopper with a delivery tube extending to the bottom of a beaker containing approximately 400 ml. of 10 percent sodium bicarbonate solution. Approximately one gm. of very finely powdered antimony metal (or iron wire or nails) is placed in the flask followed by 10 ml. of saturated sodium bicarbonate solution, the stopper being removed during the addition and then immediately replaced. The air is displaced by the carbon dioxide generated. The contents of the flask are gently boiled for 15 minutes allowing the escaping gases to bubble through the sodium bicarbonate solution. The apparatus is removed from the hot plate and allowed to cool to room temperature, leaving the sodium bicarbonate solution to establish the pressure equilibrium, carbon dioxide being generated in the flask as the sodium bicarbonate solution is drawn into the flask on cooling. When the contents of the flask are cooled the stopper is removed, 5 ml. of starch indicator solution is added, and the solution titrated to a permanent blue end point with 0.1 N iodine solution. One ml. of 0.1 N iodine is equivalent to 0.005935 grams of tin as metal.

F-4d. *Cadmium (for composition C solder).*—

F-4d(1). *Method.*—Weigh 1 gm. of the sample into a 250-ml. beaker, add 10 ml. of water, heat, and add sufficient HNO_3 (sp. gr. 1.42), a little at a time, until the alloy is decomposed. Evaporate to dryness, digest with 10 ml. of HNO_3 (sp. gr. 1.42), dilute to 50 ml. with water, bring to boiling and allow to stand, and settle for one hour keeping the temperature just below the boiling point. Filter, keeping the solution hot, through a small close-textured paper to which a little paper pulp has been added and wash with hot dilute HNO_3 (1:49). Reserve this original filtrate for cadmium.

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Transfer the paper and precipitate to a small porcelain crucible, dry and ignite. Mix the residue with 4 gm. of fusion mixture, cover with 4 gm. of the fusion mixture and heat until in quiet fusion; heat 20 minutes longer. Cool, leach with as little water as possible, remove and wash the crucible. Combine washings with the main leaching solution. The supernatant solution should be of a clear, yellow color. Filter and wash with warm (not hot) KSH wash solution. Discard the filtrate.

Dissolve the residue from fusion with 15 ml. of hot HNO_3 (1:2), wash with hot water, boil to a volume of 25 ml. and allow to stand for one hour keeping the temperature just below the boiling point. If an insoluble residue is present, filter on a small close-textured paper to which a little paper pulp has been added and wash with hot dilute HNO_3 (1:49). Dischard the tin bearing residue. Combine the filtrate with the original filtrate (from the H_2SnO_3 separation).

Boil the combined filtrates down to a volume of 100 ml., make just neutral with NH_4OH (1:3) as shown by a piece of litmus paper and then add 5 ml. of HCl (1:9) in excess. Dilute to a volume of 400 ml. with boiling hot water and allow to stand in a warm place for several hours. Filter and wash with hot water. Reserve the *filtrate*.

Dissolve the precipitate from the filter with hot HCl (1:1) and wash alternately, first with hot water and then with hot HCl (1:1). Evaporate the solution down to a volume of about 1 ml., dilute to a volume of 400 ml. with boiling hot water and allow to stand as before. Filter on a porcelain Gooch crucible and wash with hot water. Transfer the *filtrate* to a clean beaker and reserve. Discard the precipitate.

Combine the two filtrates from the BiOCl separations, evaporate to a volume of about 40 ml., add 10 ml. of H_2SO_4 (1:1), bring to boiling, cool, allow to stand 30 minutes or more, filter and wash with cold water. Discard the lead bearing residue. Dilute the filtrate to a volume of 300 ml., warm and pass H_2S for 30 minutes, allow to stand in a warm place an hour or so and again pass H_2S for 15 minutes. Filter and wash two or three times with H_2S water containing a little $(NH_4)_2SO_4$. Transfer the paper containing the precipitate back into the beaker from which the filtration was made, add 10 ml. of HNO_3 (sp. gr. 1.42) and 5 ml. of H_2SO_4 and evaporate to fumes, taking the necessary steps to remove carbonaceous matter by the addition of a drop or two of HNO_3 (sp. gr. 1.42) at the point where the solution begins to fume. Repeat the addition of HNO_3 until the solution shows no brown coloration even upon strong fuming. Cool, take up with a few ml. of water and fume. Again cool, take up with water and fume. Finally cool, dilute to a volume of 40 ml., bring to boiling, cool, allow to stand at least 2 hours, filter and wash with cold dilute H_2SO_4 (1:9). Discard the residue. Neutralize the filtrate with NH_4OH (1:2) and then make acid with 3 ml. of H_2SO_4 (1:9) in excess, dilute to a volume of 150 ml. and electrolyze overnight with a current of 1.0 ampere, using a platinum spiral for anode and a weighed platinum cylinder for cathode. Remove the cathode, rinse in water and in 95 percent alcohol, dry at 110°C. for 10 minutes, cool and weigh as cadmium plus copper.

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Dissolve the deposit from the cylinder with 5 ml. of HNO_3 (sp. gr. 1.42) and wash the electrode free from cadmium with water. Combine the electrolyte and washings and evaporate to a small volume. Wash into a small porcelain crucible and evaporate to dryness. Take up with 1 ml. of HNO_3 (1:1).

Neutralize with a saturated solution of sodium carbonate, add 4 drops of NH_4OH (sp. gr. 0.9) and titrate with standard KCN solution to the disappearance of the blue color.

Having determined the amount of copper present, deduct from the weight of copper plus cadmium. The difference is cadmium.

Ml. KCN × titre factor for standard KCN = grams of copper electroplated on cathode.

Weight of cadmium = total weight of copper and cadmium found on cathode minus weight of copper.

$$\frac{\text{Weight of cadmium}}{\text{Weight of electroplating}} \times 100 = \text{percent cadmium.}$$

F–4d(2). *Reagents required.*—

F–4d(2)a. *Fusion mixture.*—Weigh 100 gm. of Na_2CO_3, 140 gm. of K_2CO_3 and 100 gm. of sulfur into a large porcelain mortar. Grind and mix thoroughly, transfer to a glass-stoppered bottle and reserve for use.

F–4d(2)b. *Potassium hydrosulfide stock solution.* — Weigh 200 gm. of KOH into a 1000-ml. beaker, dissolve in about 800 ml. of distilled water and pass H_2S for about one hour. Make up to a volume of 2000 ml. with distilled water, mix and allow to stand overnight. Filter off portions as required for use.

F–4d(2)c. *Potassium hydrosulfide wash solution.*—Mix one part of the potassium hydrosulfide stock solution with two parts of distilled water.

F–4d(2)d. *Standard KCN solution.*—Dissolve 4 gm. of KCN in 1000 ml. of water. Standardize against pure copper as follows: Weigh approximately .040 gm. copper. Take up with a minimum amount (approximately 2 ml.) of HNO_3 (1:1) and evaporate to dryness. Neutralize with a saturated solution of sodium carbonate and continue as in the analysis. Determine the titre factor.

F–4e. *Determination of freezing point.*—Approximately 50 gm. of the alloy, or enough to completely cover the bulb of a thermometer, is placed in a small crucible and melted, avoiding heating the material higher than necessary. The metal is allowed to cool and the temperature recorded at 15-second intervals until the temperature remains constant or rises slightly and then continues to fall. The freezing point is taken as the temperature at which the thermometer remained constant or the highest temperature of the slight rise.

G. PACKAGING, PACKING, AND MARKING FOR SHIPMENT

G–1. *Packaging (for domestic and overseas).*—Solder shall be packaged in unit quantities in accordance with standard practice.

G–2. *Packing.*—

G–2a. *For domestic shipment.*—Solder, packaged as described in paragraph G–1, shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose,

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so constructed as to insure acceptance and safe delivery by common or other carrier, at the lowest rate, to the point of delivery. When specified, nailed wood boxes, wood-cleated-fiberboard boxes, wood-cleated-plywood boxes, corrugated fiber boxes, solid fiber boxes, or wirebound wood boxes used shall conform to the requirements of Federal Specifications NN–B–621, NN–B–591, NN–B–601, LLL–B–631, LLL–B–636, and NN–B–631, respectively. The gross weight of wood boxes shall not exceed approximately 200 pounds; of fiberboard boxes, approximately 70 pounds. Shipping containers shall be of uniform size and shall contain the same number of items. Solder shall be segregated as to composition and size and not more than one composition or size shall be packed in a single container.

G–2b. *For overseas shipment.*—Solder, packaged as described in paragraph G–1, shall be packed in cleated-plywood boxes, nailed wood boxes, wirebound wood boxes, fiberboard boxes, or fiberboard-lined wood boxes conforming to the requirements of Joint Army-Navy Specification JAN–P–105, JAN–P–106, JAN–B–107, JAN–P–108, and JAN–P–138, respectively. Shipping containers shall be lined with a sealed waterproof bag, or its equivalent, made from barrier-material conforming to the requirements of Joint Army-Navy Specification JAN–P–125, for case liners. The seams and closures shall be sealed with water-resistant adhesive conforming to the requirements of Joint Army-Navy Specification JAN–P–140. The gross weight of wood boxes shall not exceed approximately 200 pounds; of fiberboard boxes, approximately 70 pounds. Shipping containers shall be of uniform size and shall contain the same number of items. Solder shall be segregated as to composition, form and size and not more than one composition, form or size shall be packed in a single container.

G–3. *Marking.*—In addition to any special marking required by the contract or order, shipments for the Army shall be marked in accordance with the requirements of U. S. Army Specification 100–2; for the Navy, in accordance with the requirements of the Navy Shipment Marking Handbook.

H. NOTES

H–1. *Use.*—Solder covered by this specification is intended for application where a melting-point range of from 155° to 330°F. is required.

H–2. *Ordering information.*—Requests, requisitions, schedules, and contract or order should specify the following:

a. Title, number, and date of this specification.
b. Composition, form, and dimensions desired. (See pars. B–2, E–1, and E–2.)
c. Place of inspection. (See par. F–1a.)

H–3. Copies of National Military Establishment, Joint Army-Navy, and Federal specifications (required for Army and Air Force purchases) and U. S. Army specifications may be obtained as indicated in the "Index of United States Army, Joint Army-Navy, and Federal Specifications and Standards." Copies of this Index may be obtained from the Superintendent of Documents,

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Government Printing Office, Washington 25, D. C. Agencies within the Department of the Army, and the Department of the Air Force will obtain copies of Joint Army-Navy, U. S. Army, and Federal specifications through established departmental channels. Both the title and identifying symbol number should be stipulated when requesting copies.

H–4. Copies of National Military Establishment, Joint Army-Navy, and Federal specifications (required for Navy purchases), Navy Department specifications, and the Navy Shipment Marking Handbook may be obtained upon application to the Bureau of Supplies and Accounts, Department of the Navy, Washington 25, D. C., except that Naval activities should make application to the Supply Officer in Command, Naval Supply Center, Norfolk 11, Va. Both the title and identifying symbol number should be stipulated when requesting copies.

H–5. Copies of this National Military Establishment specification (required for Army purchases) may be obtained from the Office, Chief of Ordnance, Department of the Army, Washington 25, D. C.

Notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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